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Specification and Abstraction

Learning Objectives

After learning the concepts in this chapter, the students will be able

- To understand the concept of algorithmic problem solving.
- To apply the knowledge of algorithmic technique in problem solving.

A number of processes performed in our daily life follow the step-by-step execution of a sequence of instructions. Getting ready to school in the morning, drawing "kolams", cooking a dish, adding two numbers are examples of processes. Processes are generated by executing algorithms. In this chapter, we will see how algorithms are specified and how elements of a process are abstracted in algorithms.

6.1 Algorithms

An algorithm is a sequence of instructions to accomplish a task or solve a problem. An instruction describes an action. When the instructions are executed, a process evolves, which accomplishes the intended task or solves the given problem. We can compare an algorithm to a recipe, and the resulting process to cooking.

We are interested in executing ouralgorithms in a computer. A computer can only execute instructions in a programming language. Instructions of a computer are also known as statements. Therefore, ultimately, algorithms must be expressed using statements of a programming language.

A problem is specified by given in put data, desired output data and a relation between the input and the output data. An algorithm starts execution with the input data, executes the statements, and finishes execution with the output data. When it finishes execution, the specified relation between the input data and the output data should be satisfied. Only then has the algorithm solved the given problem.

DO YOU KNOW?

G Polya was a Hungarian mathematician. He made fundamental contributions to combinatorics, number theory, numerical analysis and probability theory. He is also noted for his work in heuristics and mathematics education, identifying systematic methods of problem solving to further discovery and invention in mathematics for students and teachers. In "How to Solve It", he suggests the following steps when solving a mathematical problem: 1. Understand the problem. 2. Devise a plan. 3. Carry out the plan. 4. Review your work.

An algorithm is a step by step sequence of statements intended to solve a problem. When executed with input data, it generates a computational process, and ends